

100 Days of Linux - Day 039

Title: Practical Assignment - Text Processing Toolkit

Today's Goal

Apply every text-processing command learned so far in one realistic data engineering workflow. No new commands are introduced today.

Why This Matters

Real projects involve cleaning, transforming, filtering, and summarizing data before it is loaded into databases or analytics platforms.

Commands Covered

sort, uniq, wc, cut, paste, tr, sed, awk

Scenario

You have received employee and sales data from different teams. Your job is to clean, inspect, and prepare the files using Linux commands.

Practical Assignment (5 Tasks)

1. Create `employees.csv` with the columns ID, Name, Department, and Salary. Add at least 8 employee records.
2. Use `cut` to display the Name and Department columns. Then use `awk` to display employees whose Salary is greater than 50000.
3. Create `departments.txt` with duplicate department names. Use `sort` and `uniq -c` to count how many times each department appears.
4. Create `first_names.txt` and `last_names.txt`. Merge them into `full_names.csv` using `paste` with a comma delimiter. Then use `wc -l` to count the number of records.
5. Create `application.log` containing INFO, WARNING, and ERROR messages. Use `grep` to find ERROR, `sed` to replace ERROR with CRITICAL (preview first, then edit in place), and `tr` to convert the final output to uppercase. Display your current working directory.

Hints

Break the assignment into small steps. Verify every result using `cat`, `head`, or `wc` before moving to the next task.

Mini Challenge

Build a `retail_project` containing `products.csv`, `orders.csv`, and `inventory.log`. Use at least five of today's commands to inspect, clean, and summarize the data.

Common Mistakes

Skipping verification after each command or editing files with `sed -i` before checking the preview.

Did You Know?

Many ETL pipelines rely on the same Linux text-processing tools to prepare raw data before loading it into Spark or databases.

Pro Tip

Combine commands with pipes (|) to build efficient one-line data-processing workflows.

Answer Key (Instructor Only)

Q1: Create employees.csv with Nano

Q2: `cut -d ',' -f2,3 employees.csv && awk -F',' '$4>50000 {print $2,$4}' employees.csv`

Q3: `sort departments.txt | uniq -c`

Q4: `paste -d ',' first_names.txt last_names.txt > full_names.csv && wc -l full_names.csv`

Q5: `grep 'ERROR' application.log && sed 's/ERROR/CRITICAL/g' application.log && sed -i 's/ERROR/CRITICAL/g' application.log && cat application.log | tr 'a-z' 'A-Z' && pwd`

Homework

Repeat the complete workflow from memory using different sample data until you can complete it without referring to the answer key.

Next Day Preview

Tomorrow you'll begin the Processes module by learning how Linux runs programs and how to inspect them with `ps` and `top`.